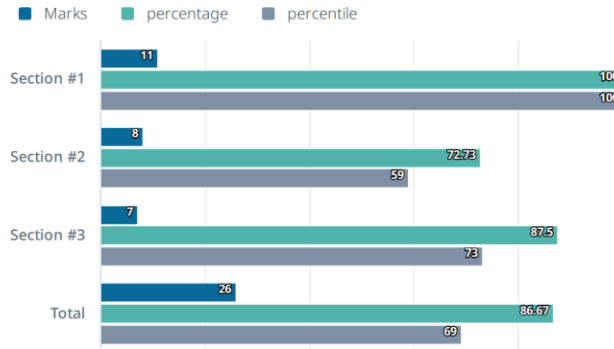


26 Marks Scored
out of 30

86.67 % 68.71 percentile
out of 147 Test
Takers

2_h 53_m 18_s Time taken
of 2hr 55mins

Marks Scored



7 of 12 Automatic Zoom

A. Scenario: We have a parallel corpus containing a simple sentence pair:
Source (English): I like coffee.
Target (Spanish): Me gusta el café.
How many possible alignments are there for this sentence pair, considering each English word can be aligned to a Spanish word or the null token ($\emptyset$)? [1 Mark]

B. Given the machine translation systems output and reference translation provided below, evaluate the machine translation system using BLEU score (Bigram) with Brevity penalty. [2 marks]
French language sentence to be translated: **Mon avion atterrira à terre**
Human translation is as follows:
Reference: My plane will land on land
Machine translation program gave the following:
Candidate: My plane will land

C. We have an NMT model trained on an English-French corpus. We are translating the English sentence "The cat sat on the mat." and are currently at the third word prediction stage (following "The" and "cat"). The model outputs the following probabilities for the next French word: [4 marks]
"est" (is): 0.3
"sur" (on): 0.4
"le" (the): 0.2
"un" (a/an): 0.1
Assume, the beam search width is set to 2 (we keep track of the top 2 most likely partial translations). Considering the current translation state and beam width, calculate the scores for all possible continuations (2-word sequences) and identify the top 2 scoring sequences.

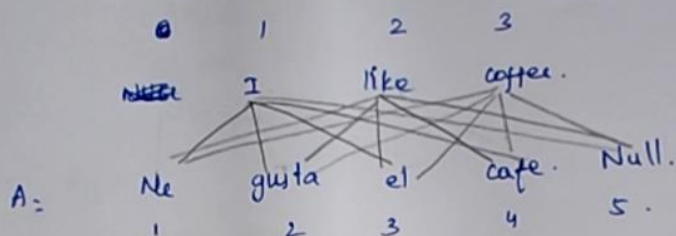
D. How will the neural machine translation approach applied for the real world application to implement machine translation for low resource Indic languages like Assamese. [2 marks]

E. Explain how transfer learning does work better for related Indian languages? [2 marks]

[1+2+4+2+2=11 Marks]

- 1) English: I like coffee.
 A) Spanish:- Me gusta el cafe.

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The possible alignment are there for this sentences pair, considering each English word can be aligned to a Spanish word or null token are 15.

$$3 \times 5 = 15 \text{ Alignment possible.}$$

There are 15 possible word Alignment for this translation.

- 1B) Machine translation system.
 French language to be translated
 Non avion atterrira à terre.

Human translation is as follows:

Ref :- My plane will land on land.

Candidate :- My plane will land.

B) BLEU Score :-

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Step 1:- Count each n-gram occurrence in candidate.

$$c(My) = 1$$

$$c(Plane) = 1$$

$$c(will) = 1$$

$$c(land) = 1$$

$$\text{Total} = 4.$$

Step 2:- Reference counts

$$R(My) = 1$$

$$R(Plane) = 1$$

$$R(will) = 1$$

$$R(land) = 2$$

$$R(on) = 1$$

^{not}
Step 3:- Clip counts.

$$\min(c(My), R(My)) = 1$$

$$\min(c(Plane), R(Plane)) = 1$$

$$\min(c(will), R(will)) = 1$$

$$\min(c(land), R(land)) = 1$$

$$\text{Total clip count} = 5.$$

$$\text{Bleu Score} = 4/4 = 1.$$

$$\text{Unigram Precision} = 1.$$

B)

BLEU Score (Bigram):

Candidate

$$C(\text{My, Plane}) = 1$$

$$C(\text{Plane, will}) = 1$$

$$C(\text{will, land}) = 1$$

$$\text{Total} = 3.$$

Ref.

$$R(\text{My, Plane}) = 1$$

$$R(\text{Plane, will}) = 1$$

$$R(\text{will, land}) = 1$$

$$R(\text{land, on}) = 1$$

$$R(\text{on, land}) = 1.$$

$$\therefore \text{Bigram Precision} = 1.$$

$$\text{BLEU Score} = \text{BP} \times \text{P}.$$

$$\text{BP} = \text{Brevity Penalty}$$

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

$$\therefore \text{BP} = e^{(1-6/4)} = e^{1-1.5} = e^{-0.5}$$

$$\text{BP} = 0.607.$$

$$\therefore \text{BLEU Score} = 1 \times 0.607$$

$$\boxed{\text{BLEU Score} = 0.607}$$

3

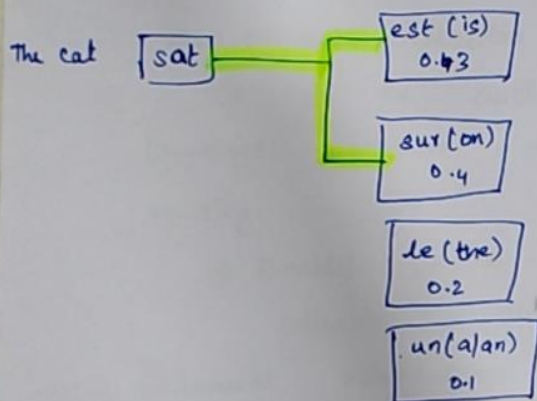
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1) c)

English : The cat sat on the mat.
Third word Prediction. "sat"

est(is) : 0.3 sur(on) : 0.4.
le(the) : 0.2 un(a/an) : 0.1

Beam search width = 2.



The next word selection from the beam search would be . est(0.3) and sur(0.4).

These two french words has the higher probability.
Therefore the sentence would be

The → cat → sat → ~~is~~ on(sur) ⇒ 0.4 Probability

The → cat → sat → is(est) ⇒ 0.3 Probability.

Other two words will be not not be selected for the next word prediction. Since the probability is low compared to the others.

(4)

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(5)

1D) Neural Machine Translation:- (Assamese)

It is the automated translation of language native to the Indian subcontinent like the assamese. Translation of one language to the other Indian language using neural network called as Neural Machine Translation.

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Machine Translation for low resource Indian languages:-

Step 1:- Data Collection & Curation:-

It is the crucial step for translating the low resource language because the resource (data) won't be available readily.

a) Parallel Corpora Development

collecting & curating parallel text from web, government publication and other resources.

b) Crowd/Sourcing:-

Platforms like wikipedia, specialized - projects can leverage crowdsourcing to generate parallel corpora.

c) Data Augmentation:-

- 1) Synthetic Data generation.
- 2) Cross lingual Transfer.

d) Leveraging Existing Source.

- 1) Open source & shared dataset
- 2) Public and Government source.

e) Neural Machine Translation:-

The neural machine translation models are more robust to noise and can learn efficiently from



limited data available.

* Once we have limited data.

we can feed the input to the deep learning models like the transformer, RNN, LSTM etc. to train the model to generate the required translation.

* NMT uses unsupervised learning techniques. So the low resource Indic language like assamese can be easily trained and translated to other languages.

1E) Transfer Learning for Indian Languages:-

Transfer Learning can be done based on following two approaches for the Related languages.

1) Using Linguistic Similarities

Based on the similarities among each language it can be learned and trained based on these similar linguistic languages.

2) Shared Vocabularies:-

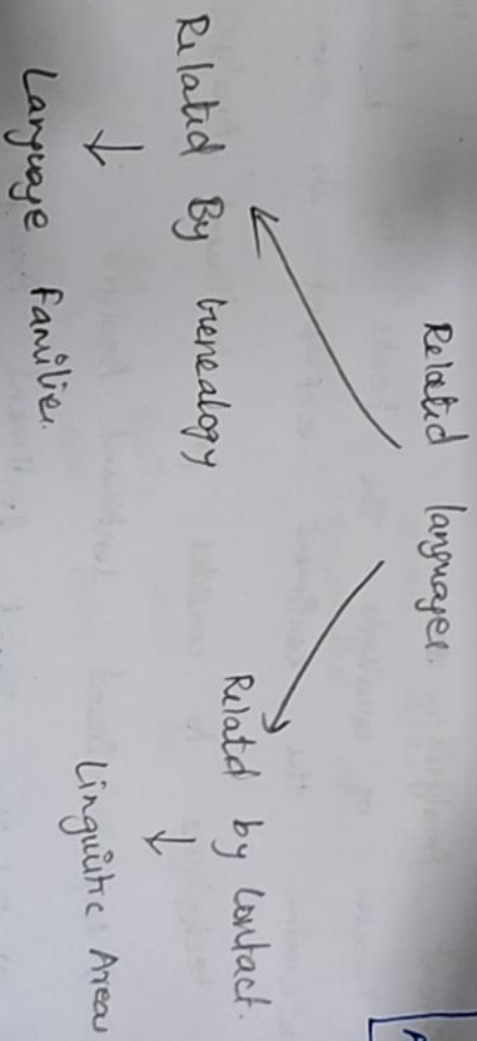
Transfer Learning will be effective for the languages within same family or with significant similar linguistic contexts, enabling efficient scaling of translation capabilities across multiple languages with limited data.

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Automatic Zoom

A. Discuss the symbiotic relationship between Information Extraction (IE) and Knowledge Graphs (KGs). Can IE assist in the creation and enrichment of Knowledge Graphs, or does the existence of Knowledge Graphs enhance the performance of Information Extraction systems? Provide arguments for both perspectives and support your answer with examples. **[3 Marks]**

B. Given the following sentences:

"Apple Inc. is headquartered in Cupertino, California."
"Tim Cook is the CEO of Apple Inc."
"Apple Inc. announced the launch of its new iPhone model."
"The event will take place at the Steve Jobs Theater."

Extract the named entities, relations, and events mentioned in these sentences. Provide a brief description of each named entity, relation, and event extracted from the sentences. **[4 Marks]**

C. Consider an ACE model trained for Named Entity Recognition (NER) tasks on text data. The model undergoes evaluation on a test set consisting of 100 instances. Out of these instances, the model correctly identifies 70 instances as Named Entities (NEs), incorrectly identifies 10 instances as NEs, and misses 20 instances that are actually NEs. Calculate the precision, recall, F1-score, and accuracy for this ACE model. Provide your calculations and interpretations. **[4 Marks]**

[3+4+4=11 Marks]

2)
A)

Information Extraction vs knowledge Graph:-

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Identifying specific pieces of information and data in an unstructured or semi structured textual document. is called Information extraction.

Both the knowledge Graph and the Information Extraction identify the information from unstructured format and converts it to the structured format.

IE ^{with} KG to enhance its performance.

The Information Extraction is the key process for extracting information from any source. The knowledge graph converts the extracted information and stores it in the easier accessible format.

* The knowledge graph is very effectively accessed and the information retrieval is very efficiently done compared to IE.

* But the knowledge graph has very limited information from the data that are available. So the Information Extraction technique can enhance the process of building the structured graph format and can be stored in the knowledge graph.

eg:- Google Search engine using knowledge graph information which can fetch information in milliseconds for Internet search.

2) B.ORG I.ORG

B). "Apple Inc is headquartered in Cupertino,
B LOC

California".

I LOC

"Tim Cook is the CEO of Apple Inc."
B PER. I PER B.ORG I.ORG.

"Apple Inc. announced the launch of its New iPhone model."
B.ORG I.ORG.

"The event will take place at the Steve Jobs Theater".
B LOC I LOC I LOC.

Named Entities:-

Apple Inc - Organisation.

Tim Cook - Person.

Cupertino - Location.

California - Location.

Steve Jobs Theatre - Location.

NER identifies and classifies the entities into person, organisation, location, date etc. like the above. ~~named~~ ^{ti} named entities.

Relation Extraction:-

After identifying the entities then the relationship between the entities are extracted called as Relation Extraction.

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(Tim Cook, CEO, Apple Inc.)

(Apple Inc, headquarters, California).

(Apple Inc, launched, iPhone)

(iPhone, launched, Steve Jobs Theatre).

~~Event~~ ~~Extraction~~:-

Event Extraction:-

iPhone launch will take place at Steve Jobs Theatre.
No temporal event specified.
^

2)

4) Total test instance = 100.

Correctly classified = 70.

Incorrectly classified = 10.

Missed Instance = 20.

$$1) \text{ Precision} = \frac{TP}{TP+FP}$$

$$= \frac{70}{70+10}$$

$$= \frac{70}{80} = \frac{7}{8}$$

$$\text{Precision} = 0.875$$

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$$7) \text{ Recall} = \frac{TP}{TP + FN}$$

$$= \frac{70}{980} = 7.1\%$$

$$\text{Recall} = 7.1\%$$

$$3) \text{ F1 Score} = \frac{2 \times (\text{Prec} \times \text{Recall})}{\text{Prec} + \text{Recall}}$$

$$= \frac{2 \times (0.875 \times 0.071)}{(0.875 + 0.071)}$$

$$= 0.109 \times 2$$

$$\text{F1 Score} = 0.218$$

$$4) \text{ Classification Accuracy} = \frac{70}{100} = 0.7$$

3)

A) Sentiment Analysis:-

Process of analysis the texts to determine the sentiment expressed in them.

Two techniques to consider for Sentiment Analysis:-

- 1) Aspect Based Sentiment Analysis
- 2) Polarity Based Sentiment Analysis

Aspect Based:-

The sentiment analysis is analysed based on the content of specific aspects or features of a subject.

Like the quintuple (e, a, s, h, t). technique.

- 1) e (Entity) - Main subject of the opinion.
- 2) a (Aspect) :- A feature or part of the entity.
- 3) s (Sentiment) :- Positive, Negative, Neutral. about the aspect.
- 4) h (holder) :- The person or entity expressing the opinion.
- 5) t (Time) :- Time at which the opinion was expressed.

eg:- "The camera takes great photos, but battery life is short."

Aspects of the Phone {
camera → positive
battery life → negative

h - User.
E → Phone.
A → Camera
s → positive
t → 11/24 1:00pm.

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Advantages:-

* Able to get aspect level feature and the sentiment expressed about that particular aspect.

* Can be easily able to evaluate the product or entity that are being analysed.

Aspect Polarity Based Sentiment Analysis:-

The sentiment is classified into categories like positive negative and ~~neutral~~ neutral.

Turney Algorithm:-

* Focuses on the semantic orientation. (positive or negative sentiment).

* Extract phrases from the text which tries to find the pattern like adj followed by a noun.

eg. very good
adj NOUN
↓
positive

not good
adj NOUN
↓
negative.

Advantages:-

1) Simple to implement and able to get the content of the particular sentence.

Comparison with Aspect Polarity vs Polarity:-

* The aspect based analysis will be more effectively compared to the polarity based. because in the same sentence they will be

(10)
talking about multiple topic at the
single blog or post.

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* We have to analyse based on the different
aspects of the review or the post.

* Polarity based will be good at the word
or sentences level sentiment analysis. but if they
are talking about multiple thing in the same
sentence then we should proceed for aspect based
sentiment analysis.

3B):- Target Dependent LSTM:-

The target dependent LSTM is the extension
of the LSTM model. The target is set and
the deep neural network model with the LSTM
will be able to perform extremely well for
that particular target.

Applications of TD-LSTM:-

For the Target and Aspect Based sentiment
analysis it would be an ideal choice. The sentiment
toward the target, its aspects and TD-LSTM
will classify the sentiment powerfully.

"The book is a hardcover version, but the price is a
bit high".

Target
Book
Book

Aspect
Cover.
Price

Emotion:
Positive
Negative

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* Specific aspects that is targeted from the blog or post or website then TD-LSTM will fetch the target and classify into the right category.

Advantages of TD-LSTM over traditional LSTM:-

The TD-LSTM will perform well on the ~~target~~ ^{aspect} based sentiment analysis over the traditional LSTM model.

Since the target dependent LSTM has an additional feature as the target it will be able to train the model based on the specific target and the long range dependency of the particular target will be kept for the entire content.

In the traditional LSTM architecture, since the target is not set over the period of time it will forget and not work well for all scenarios.

Real World Applications:-

- 1) Chat Bot ~~based~~ ^{for} on the document analysis
- 2) Question Answering bot with the specific target product.
- 3) Blogs or post Analysis.